

FL4

(i)  $X$  compact topological space.

$$C(X) = C(X, \mathbb{R}) \quad \text{with } \|\cdot\|_\infty$$

Metric topology,  $d(f, g) = \|f - g\|_\infty$ .

$(C(X), \|\cdot\|_\infty)$  is a Banach space.

Let  $K = C(X)$ . TFAE

(i)  $K$  is compact

(ii)  $K$  is sequentially compact

(iii)  $K$  is closed and totally bounded.

② Let A subset (family)  $\mathcal{F} \subset C(X)$  is

(i) precompact if  $\overline{\mathcal{F}}$  is compact

(ii) equicontinuous if  $\forall x \in X, \forall \varepsilon > 0$

$\exists$  nbh  $U$  of  $x$  :  $f(U) \subset B(p(x); \varepsilon), \forall f \in \mathcal{F}$

(i.e.  $y \in U \Rightarrow |f(y) - f(x)| < \varepsilon$ ),  $\forall f \in \mathcal{F}$

(iii) bounded if  $\exists \mathcal{F} \subset B(0; C)$  large enough  $C$ , i.e.  $\|f\|_{\infty} \leq C \quad \forall f \in \mathcal{F}$ .

Ex  $f_n : [-1, 1] \rightarrow [-1, 1], x \mapsto \frac{1}{(1+x^2)^n}$

Not precompact:  $f_n \rightarrow \begin{cases} 1, & x=0 \\ 0, & x \neq 0 \end{cases}$

Bounded:  $\|f_n\|_{\infty} \leq 1 \quad \forall n$ .

Not equicontinuous at  $x=0$ : fix  $0 < \varepsilon < 1$

Let  $\delta > 0$ , suppose  $f_n(B(0; \delta)) \subset B(1; \varepsilon) \quad \forall n$

Pick  $N$  s.t.  $\frac{1}{(1+\delta^2)^N} < \frac{1-\varepsilon}{2}$  Contradiction.  $\square$

③ Arzelà-Ascoli Theorem Let  $X$  compact.

A family  $\mathcal{F} \subset C(X)$  is precompact

$\Leftrightarrow$  it is uniformly bounded and equicontinuous.

Lemma Let  $\{f_n\}_{n=1}^{\infty}$  uniformly bounded equicontinuous sequence in  $C(X)$ .

Then there is a uniformly convergent subsequence.

Proof For each  $x \in X$  and  $n \in \mathbb{Z}_+$ , pick an open nbhd  $U_{x,n}$  of  $x$  s.t.

$$y \in U_{x,n} \Rightarrow |f_k(x) - f_k(y)| < \frac{1}{n} \quad \forall k \in \mathbb{Z}_+.$$

For fixed  $n$ , the  $U_{x,n}$  ( $x \in X$ ) cover  $X$ .

Subcover  $U_{x_{n1}, n} \cup \dots \cup U_{x_{nm_n}, n}$ .

④ Rename the  $(x_{nj})_{j=1, n=1}^{m, \infty}$  as  $(y_l)_{l=1}^{\infty}$ .

We use the usual diagonal procedure to extract a subsequence of the  $f_n$ 's

which converges at each  $y_l$ :

1) The real sequence  $(f_n(y_1))_{n=1}^{\infty}$  is bounded.

Pick a convergent subsequence  $(f_{n_1}(y_1))_{k=1}^{\infty}$ .

2) The sequence  $(f_{n_1}(y_2))_{k=1}^{\infty}$  is bounded

Pick a convergent subsequence  $(f_{n_2}(y_2))_{k=1}^{\infty}$ .

(Then  $f_{n_2}(y)$  converges as  $n \rightarrow \infty$  at  $y=y_1$  and

$y=y_2$ .)

3) Repeat.

The diagonal sequence  $f_{n_k}(y)$  converges

at each  $y=y_l$ .



⑤ Let us go back to the notation  $(x_{nj})$  instead of  $(y_\ell)$  and write  $f_{k_1}, f_{k_2}, \dots$  for the subsequence  $f_{k_k}$ .

Since the sequence  $(f_n)$  is equicontinuous there is for each  $x_{nj}$  a neighbourhood  $W_{nj}$  such that

$$y \in W_{nj} \Rightarrow |f_k(x_{nj}) - f_k(y)| < \frac{1}{n} \quad \forall k \geq 1.$$

Fix  $x_{nj}$ . Since  $(f_{k_\ell}(x_{nj}))_{\ell=1}^\infty$  converges it is a Cauchy sequence.

$$\text{So } \exists N : \ell, m \geq N \Rightarrow |f_{k_\ell}(x_{nj}) - f_{k_m}(x_{nj})| < \frac{1}{n}$$

Thus for each  $y \in W_{nj}$ ,  $\ell, m \geq N$ :

$$|f_{k_\ell}(y) - f_{k_m}(y)| < \frac{3}{n}$$

But  $(W_{nj})_{j=1}^{\infty}$  covers  $X$

$$\Rightarrow \|f_{k_\ell} - f_{k_m}\|_\infty < \frac{3}{n},$$

i.e.  $(f_{k_\ell})_{\ell=1}^\infty$  is uniformly Cauchy, so uniformly convergent.  $\square$

## ⑥ Proof of Arzela-Ascoli

$\Leftarrow$  If  $\mathcal{F}$  is uniformly bounded and equicontinuous, the lemma shows that each sequence in  $\mathcal{F}$  has a uniformly convergent subsequence. Hence  $\overline{\mathcal{F}}$  is (sequentially) compact.

$\Rightarrow$  If  $\mathcal{F}$  is precompact then  $\overline{\mathcal{F}}$  is totally bounded.

I.e. for each  $\varepsilon > 0$  there are  $f_1, \dots, f_n \in \mathcal{F}$

s.t.  $\overline{\mathcal{F}} \subset B(f_1; \varepsilon) \cup \dots \cup B(f_n; \varepsilon)$ .

For each  $x \in X$  pick a nbh  $U_x$  s.t.

$$f_k(U_x) \subset B(f_k(x); \varepsilon), \quad k = 1, \dots, n.$$

If  $f \in \mathcal{F}$  then  $y \in U_x \Rightarrow |f(x) - f(y)| < 3\varepsilon$

so  $\mathcal{F}$  is equicontinuous.  $\square$

## ⑦ Peano space-filling curve

Def Let  $X$  a normed space.

A function  $f: [a, b] \rightarrow X$  is said

to satisfy a Lipschitz condition of

exponent  $\alpha$  ( $0 < \alpha \leq 1$ ) if

$$\|f(s) - f(t)\| \leq M |s - t|^\alpha, \quad \forall s, t \in [a, b]$$

where  $M$  is some constant.

⑧ Theorem There exists a function

$$P: [0,1] \rightarrow [0,1] \times [0,1] \text{ s.t.}$$

(i)  $P$  maps  $[0,1]$  surjectively and continuously onto  $[0,1] \times [0,1]$

(ii)  $P$  is Lip- $\frac{1}{2}$ , i.e.,

$$\|P(s) - P(t)\| \leq M |s - t|^{\frac{1}{2}}, \quad 0 \leq s, t \leq 1$$

(iii)  $P$  is measure-preserving:

$$\forall 0 \leq a < b \leq 1: \text{Area}(\{P(t); a \leq t \leq b\}) = b - a.$$

We sketch the proof of (i) + (ii).

Remark To prove surjectivity it suffices

to show that  $f$  is continuous and

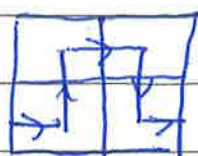
$f([0,1])$  is dense in  $[0,1] \times [0,1]$ .

This is because  $f([0,1])$  is compact

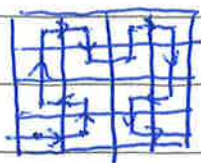
(hence closed) in  $[0,1] \times [0,1]$ .



# ⑨ Hilbert's iterative procedure.



$P_1(H)$



$P_2(H)$

Quartic intervals Subdivide  $[0,1]$  in powers of  $\frac{1}{4}$ .

First generation:  $I_1 = [0, \frac{1}{4}]$ ,  $I_2 = [\frac{1}{4}, \frac{1}{2}]$ ,  $I_3 = [\frac{1}{2}, \frac{3}{4}]$   
 $I_4 = [\frac{3}{4}, 1]$

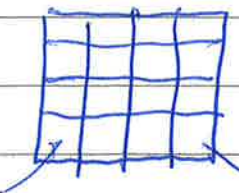
$k$ :th generation  $I_\ell = [\frac{\ell}{4^k}, \frac{\ell+1}{4^k}]$ ,  $\ell = 0, \dots, 4^k - 1$ .

A chain is a decreasing sequence

$I^1 \supset I^2 \supset I^3 \supset \dots$  where  $I^k$  is of  $k$ :th gen.

Dyadic squares in  $[0,1] \times [0,1]$  are obtained

by successively bisecting sides,

First gen  Second  etc.

A chain  $S^1 \supset S^2 \supset S^3 \supset \dots$ ,  $S^k$  of  $k$ :th generation.

(10) Def A dyadic correspondence

$\Phi$ : quateric intervals  $\rightarrow$  dyadic squares.

(i)  $\Phi$  is bijective

(ii) respects generations:  $\Phi(I^k) = S^k$

(iii) respects inclusions:  $I^m \subset I^k \Rightarrow \Phi(I^m) \subset \Phi(I^k)$

Lemma There is a unique dyadic correspondence

st. (i) If  $I, J$  are adjacent intervals of the same generation then  $\Phi(I), \Phi(J)$  are adjacent squares.

(ii) In generation  $k$ , if  $I_-$  is the leftmost interval and  $I_+$  the rightmost then  $\Phi(I_-)$  = lower left corner

$\Phi(I_+)$  = lower right corner.



Proof Induction: Stein-Shakarchi, Sec 7.3.  $\square$

⑪ Let  $\Phi$  be the dyadic correspondence provided by the lemma.

Write  $I$  with generators as  $I_1, \dots, I_{4^h}$

$S_1, \dots, S_{4^h},$

where  $\Phi(I_j) = S_j$

Let  $\begin{cases} t_j = \text{mid-point of } I_j \\ x_j = \text{center of the square } S_j \end{cases}$

Define a piecewise-linear function

$P_h : [0, 1] \rightarrow [0, 1] \times [0, 1]$  as follows:

1)  $P_h(t_j) = x_j, \quad 1 \leq j \leq 4^h$

2)  $t_0 := 0, x_0 := (0, 2^{-h-1}), P(t_0) = x_0$

3)  $t_{4^h+1} := 1, x_{4^h+1} := (1, 2^{-h-1}), P(t_{4^h+1}) = x_{4^h+1}$

4) Extend by linearity to each  $[t_j, t_{j+1}]$



(12) Note that  $\|x_j - x_{j+1}\| = 2^{-k}$  while

$$|t_j - t_{j+1}| = 4^{-k}$$

$$\text{So } \|P'_k(t)\| = 4^k \cdot 2^{-k} = 2^k \text{ except at } t = t_j$$

$$\Rightarrow \|P_k(t) - P_k(s)\| \leq 2^k |t - s| \quad (*)$$

But for each  $t \in [0, 1]$ , the points

$P_k(t), P_{k+1}(t)$  belongs to the same

dyadic square of gen  $k$ , which

has diameter  $\sqrt{2} \cdot 2^{-k}$

$$\Rightarrow \|P_k - P_{k+1}\|_\infty \leq \sqrt{2} \cdot 2^{-k}$$

Hence the limit

$$P(t) := \lim_{k \rightarrow \infty} P_k(t) = P_1(t) + \sum_{j=1}^{\infty} (P_{j+1}(t) - P_j(t))$$

exists with uniform convergence.

$P(t)$  is continuous and  $P([0, 1])$

is clearly dense in  $[0, 1] \times [0, 1]$ .



(13) We finally prove  $P(t)$  is Lip  $-\frac{1}{2}$ .

For each  $j$ ,

$$\begin{aligned}\|P(t) - P_j(t)\| &= \sum_{k=j}^{\infty} \|P_{k+1}(t) - P_k(t)\| \leq \\ &\leq \sqrt{2} \sum_{k=j}^{\infty} 2^{-k} = \sqrt{2} \cdot 2^{-j}\end{aligned}$$

It follows that

$$\begin{aligned}\|P(s) - P(t)\| &\leq \|P_j(s) - P_j(t)\| + \|(P - P_j)(s)\| + \|(P - P_j)(t)\| \\ &\stackrel{(*)}{\leq} 2^j |t-s| + 2\sqrt{2} \cdot 2^{-j}, \quad j=1, 2, \dots\end{aligned}$$

For given  $s, t$ , now choose  $j$  s.t.

$$4^j |t-s| \leq 1 \quad \text{and} \quad 1 \leq 4^{j+1} |t-s|.$$

Then  $2^j |t-s| \leq 2^{-j}$  and  $1 \leq 2 \cdot 2^j |t-s|^{\frac{1}{2}}$

$$\text{Hence } \|P(s) - P(t)\| \leq (1 + 2\sqrt{2}) \cdot 2^{-j}$$

$$\leq 2(1 + 2\sqrt{2}) \cdot |t-s|^{\frac{1}{2}}. \quad \square$$